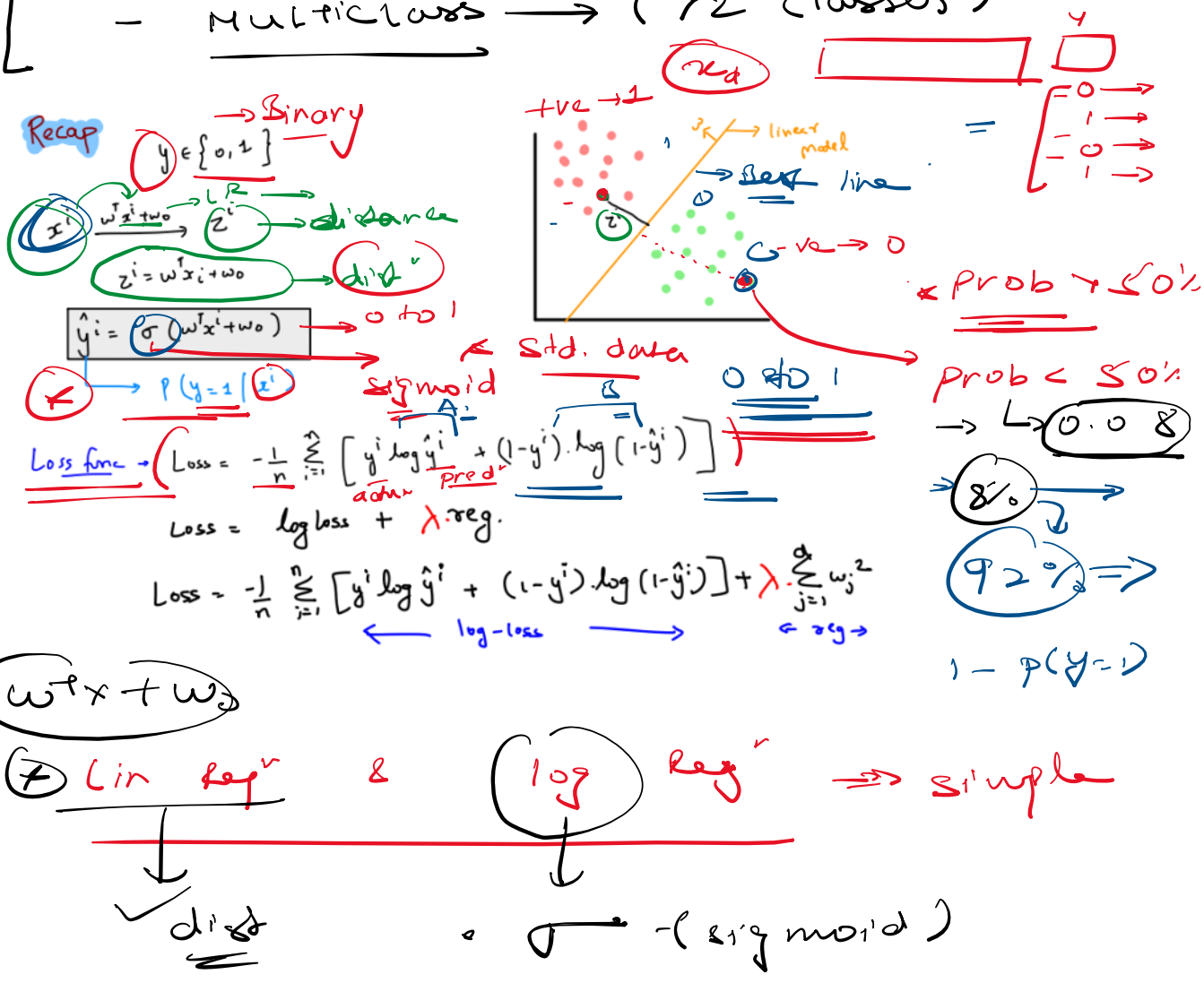
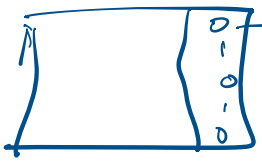
Logistic Regression – 2

Agenda

- Recap
- Derivation (maths)
- log odds
- outliers
- Multiclass → (>2 classes)



DATA → $y = 0$ ⇒ B



0 or 1

* Loss func^n → log loss

* Regularization ⇒
  → should always be applied

Lin Reg^n
- Lasso
- Ridge → $\lambda = 0$ ⇒ Lin Reg^n

= Lin Reg^n → Ridge

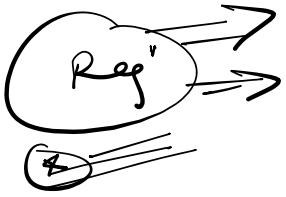
*

loss = log loss + λ Reg^n

Part of ML

λ → tuner

Reg^n constant



* updated loss func^n ⇒

$$loss = -\frac{1}{n}\sum_{i=1}^{n}\left[y^i \log \hat{y}^i + (1-y^i)\log(1-\hat{y}^i)\right]$$

$$loss = -\frac{1}{n}\sum_{i=1}^{n}\left[y^i \log \hat{y}^i + (1-y^i)\log(1-\hat{y}^i)\right] + \lambda \sum_{j=1}^{d} w_j^2$$

flexible obj func

argmin w, w_0

$x = 0$

Derivative

GD → grad. → update wts

**

$$loss = -\frac{1}{n}\sum_{i=1}^{n}\left[y^i \log \hat{y}^i + (1-y^i)\log(1-\hat{y}^i)\right] + \lambda \sum_{j=1}^{d} w_j^2$$

$$\Rightarrow 2\lambda \sum_{j=1}^{d} w_j$$

J

1 DATA POINT ⇒

$\rightarrow \sigma(w^T x + w_0)$

$$L = -\left[y \cdot \log \hat{y} + (1-y) \cdot \log(1-\hat{y}) \right]$$

$\hat{y} = \sigma(w^T x + w_0)$

↳ pred^n

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

* Sigmoid property → n/w

(*) $$\frac{d\,\sigma(z)}{dz} = \sigma(z) \cdot (1 - \sigma(z))$$ ⟺ property

(←)

$$\frac{d}{dz}\left(\frac{1}{1+e^{-z}}\right)$$

$\hat{y} = \sigma(z)$

(←) $$\frac{d\hat{y}}{dz} = \hat{y} \cdot (1-\hat{y})$$ ⇒ ✓

(**)

(*) $L = -\left[y \cdot \log \hat{y} + (1-y) \cdot \log(1-\hat{y}) \right]$ where $\hat{y} = \sigma(w^T x + w_0)$

(*)

$$L = -\left[y \cdot \log \hat{y} + (1-y) \cdot \log(1-\hat{y}) \right]$$

removing

(A) (B)

$\hat{y} = \sigma(w_1 x_1 + w_2 x_2 + \dots + w_j x_j + \dots + w_n x_n)$

z

w_j

simple add^{n} Rule $\Rightarrow \frac{dz}{dw_j}$ (x_j)

$$\frac{\partial L}{\partial w_j} = \frac{\partial L_A}{\partial w_j} + \frac{\partial L_B}{\partial w_j}$$

(*)

$$\frac{\partial L_A}{\partial w_j} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z} \cdot \frac{\partial z}{\partial w_j}$$

$\Rightarrow$ Chain Rule

$\frac{\partial}{\partial x} \log x = \frac{1}{x}$

$\frac{y}{\hat{y}} \quad \hat{y}(1-\hat{y}) \cdot x_j$

$\Rightarrow y(1-\hat{y}) \cdot x_j$

Point A

Part A

$$\frac{\partial L_B}{\partial w_j} = \frac{\partial L_B}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z} \cdot \frac{\partial z}{\partial w_j}$$

$$\frac{-(1-y)}{(1-\hat{y})} \; \hat{y}(1-\hat{y}) \cdot x_j$$

$$= -(1-y)\,\hat{y}\,x_j$$

Part B

$$\frac{\partial L_A}{\partial w_j} + \frac{\partial L_B}{\partial w_j}$$

$$= y \cdot (1-\hat{y}) \cdot x_j - (1-y)\,\hat{y} \cdot x_j$$

$$= [y - y\hat{y} - \hat{y} + y\hat{y}]\, x_j$$

$$= -[y - \hat{y}]\, x_j$$

$$\Rightarrow [\hat{y} - y] \cdot x_j$$

1 DP

$$\frac{\partial L}{\partial w_j} = \frac{1}{n} \sum_{i=1}^{n} (\hat{y}^i - y^i)\, x_j^{(i)}$$

ALL DP

Avg loss

1 DP → no of derivatives

+ Reg derivative

Cin Reg

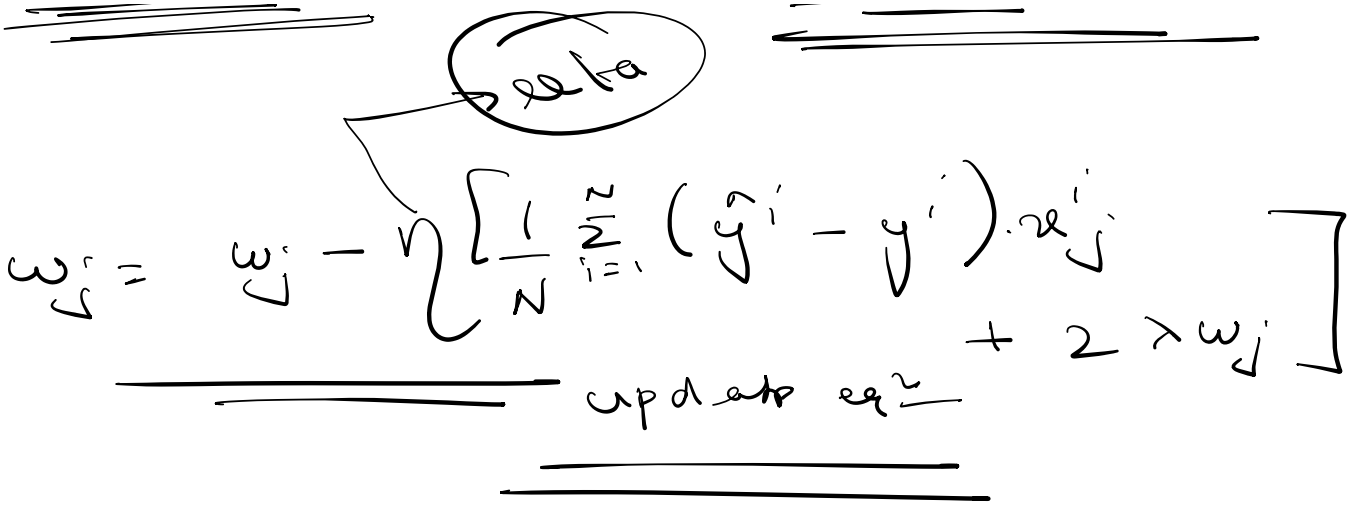
Regression

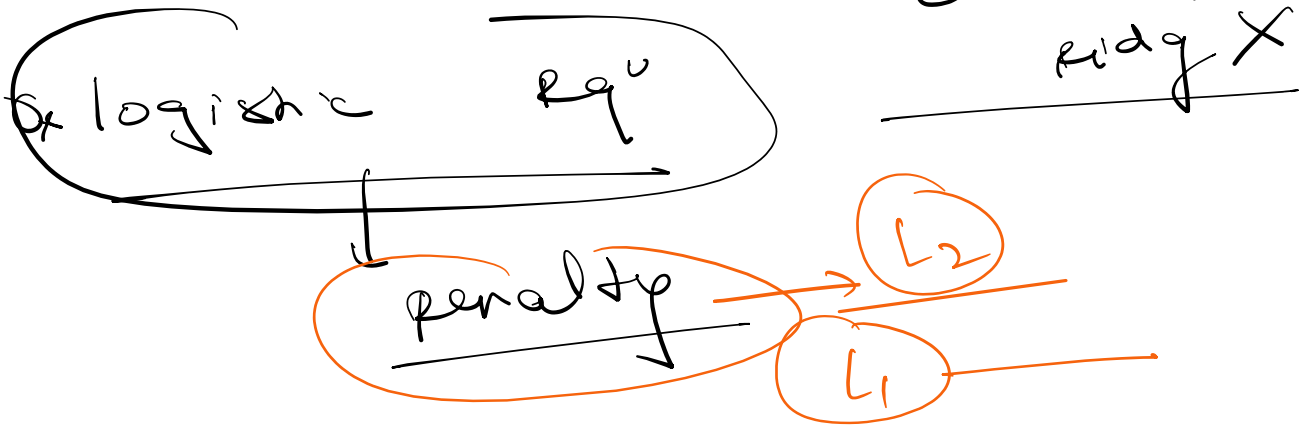
$y_i \rightarrow R$

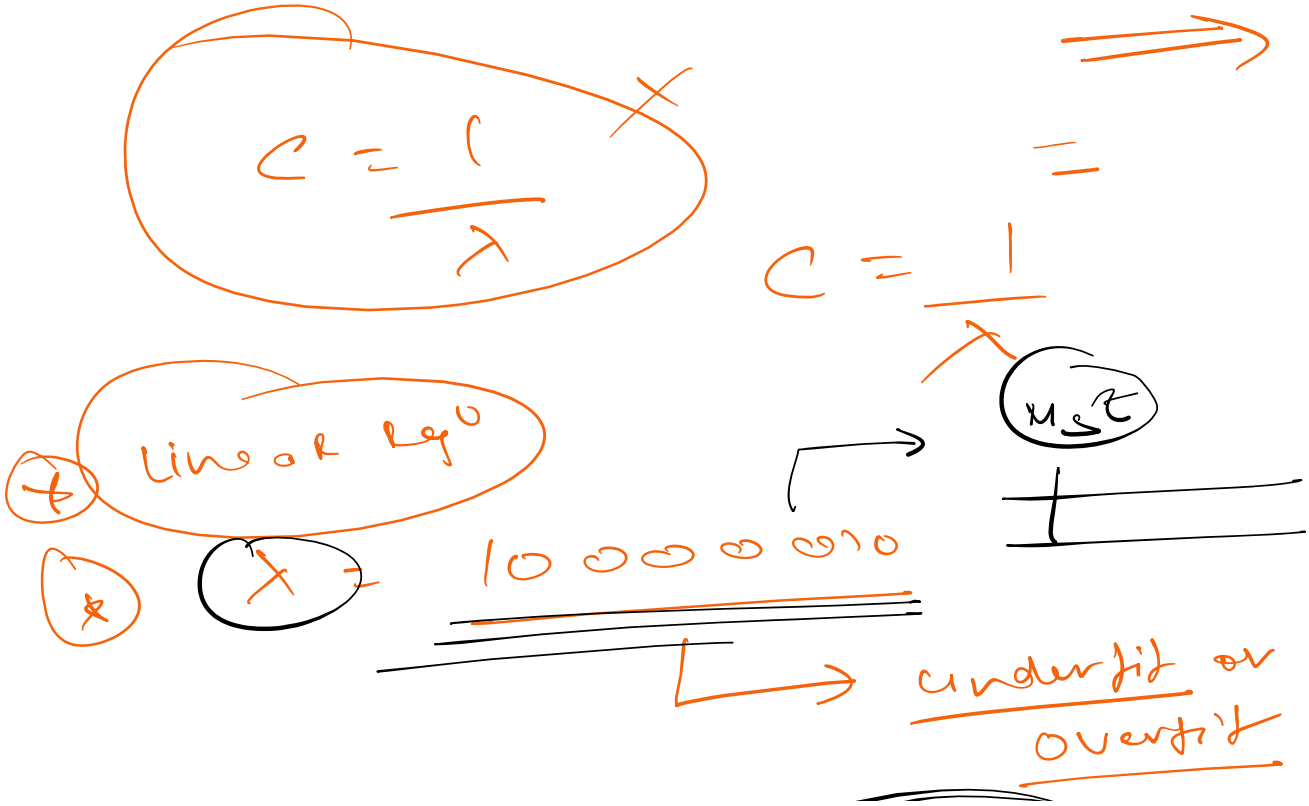
Log Reg

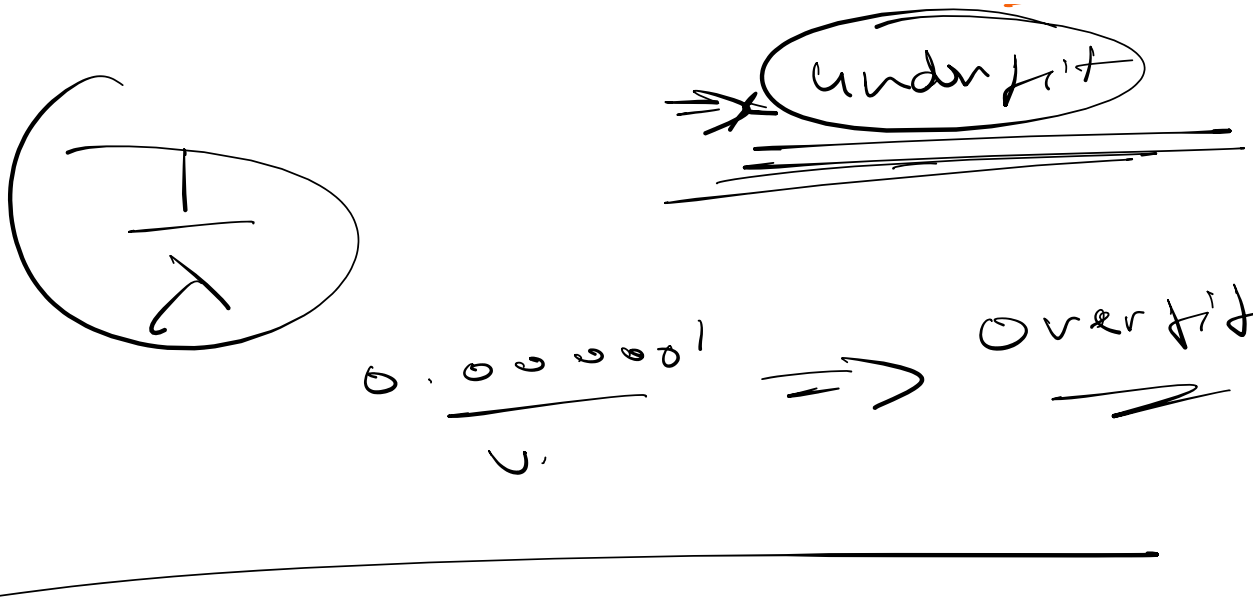
Class

$y_i \in (0, 1)$


delta
$w_j = w_j - \eta \left[\frac{1}{N} \sum_{i=1}^{N} (\hat{y}^i - y^i) \cdot x_j^i + 2\lambda w_j \right]$
update eq^n


Lasso X
Ridge X
& logistic Reg^n
penalty
L_2
L_1
Elastic —


$C = \frac{1}{\lambda}$ X
=
$C = \frac{1}{\lambda}$
MST
Linear Reg^n
$\lambda = 10000000$
underfit or overfit


underfit
overfit
earlier
λ = small → overfits
C = large → overfits
Ridge
λ ↑ = C ↓
λ ↓ = C ↑
min
10:10
Quiz time!
Time Left: 11s
What is the effect of increasing the regularization rate (C) in logistic regression?
5 users have participated
A The model becomes less prone to overfitting 40%
B The model's training accuracy increases 0%
C The model becomes more prone to overfitting 60%
D The model's test accuracy increases 0%
End Quiz Now
C ↑
λ ↓
λ = 0

MSE

Overfitting

other aspects of Logistic Reg

Odds => Prob

=> odds of India winning AFG

: 4 : 1

Ind vs Aft

5

Radio

odd Ratio = Prob (winning) → 4
Prob (loosing) → 1

x

4 : 1

4/5 / 1/5 = 4/1 = 4 : 1

Actual

Ind winning = 4/(1+4)

Aft = 1/(1+4)

odds → Ratio of winning vs loosing

AFG ki India => 1 : 4

1 : 4

$$\text{odds} = \frac{p}{1-p}$$

→ P(winning)

→ P(loosing)

log - odds → $\frac{p}{1-p}$

$$p = P(y=1 \mid x) = \sigma(z)$$

$$p = \frac{1}{1+e^{-z}}$$

$$= \frac{1}{1+\frac{1}{e^z}} \Rightarrow \frac{\frac{1}{e^z+1}}{\frac{e^z}{e^z+1}} = \frac{e^z}{e^z+1}$$

$$\sigma(z) = \frac{1}{1+e^{-z}} \quad , \quad \sigma(z) = \frac{e^z}{e^z+1}$$

more popular — same — less popular

$$1 - p = 1 - \frac{e^z}{e^z+1} = \frac{1}{e^z+1}$$

e^z

$z = w^T x + w_0$

x logits

$$\text{odds} = \frac{p}{1-p} = \frac{\frac{e^z}{e^z+1}}{\frac{1}{e^z+1}} = e^z$$

$$\log\left(\frac{p}{1-p}\right) = \log(e^z)$$

INTERVIEW

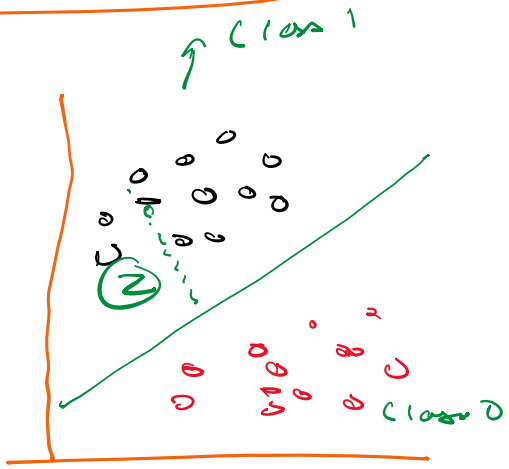
$$\log \text{odds} = z$$

logits

$w^T x + w_0$

log odds = logits =

dist ↑ → log odds ↑



$\sigma(z)$ —

Quiz time!

Time Left: 17s

How are log odds transformed into probabilities in logistic regression?

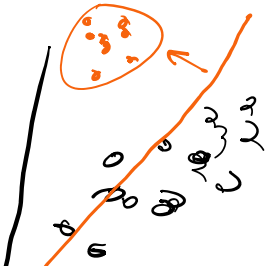
5 users have participated

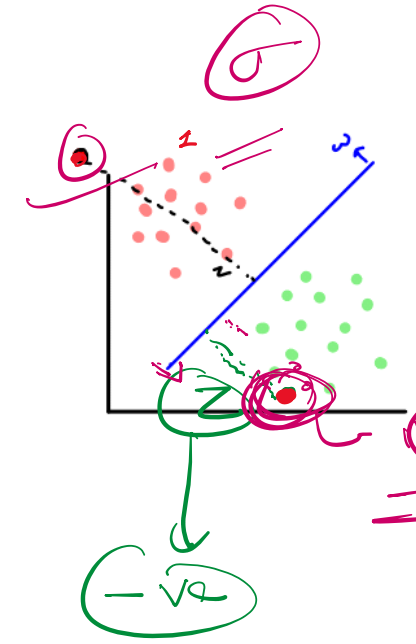
A	By applying the sigmoid function	20%
B	By taking the exponential function	60%
C	By dividing by the odds ratio	20%
D	By subtracting the intercept term	0%

End Quiz Now

impact of OUTLIERS

Linear Reg





Case 1

→ outlier on correct side

$y = 1$

$Loss = -[\log \hat{y}]$

$\hat{y} = \sigma(z)$ → huge dist

$= 0.9999$

$\hat{y} \approx 1$, $\log(1) = 0$

$Loss = -\log(1) = 0$

Loss = 0 ✗

case 2 ⇒ outlier is on opposite side

$-y \log \hat{y}$

(-ve value)

σ(-ve value)

$L = -\log \hat{y}$ (0 to 1)

$z \rightarrow$ -ve

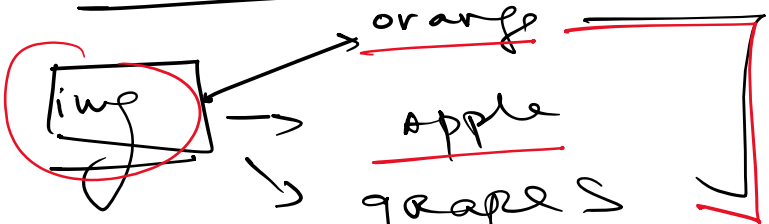
$\sigma(z) = \hat{y} =$ +ve $=$ 0.02

$-\log(0.02)$

$-(-3.19) =$ 3.19

large +ve loss

Multiclass classification



3

OVR

One Vs Rest

One Vs All

3

> 2 classes

One Vs All

3 classes ⇒ 3 log Reg^r models

n ⟶ # of DP

d ⟶ # of features

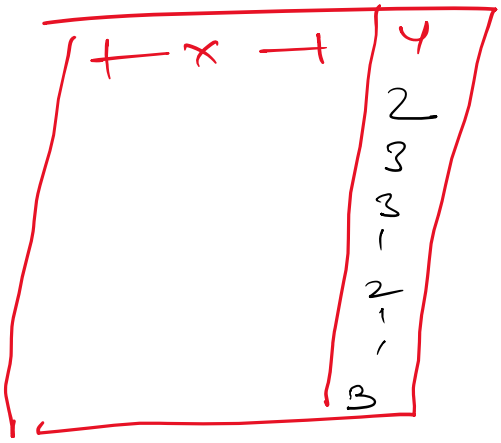
3 models

3 models

M_1 → orange (1) or not (0) → Binary

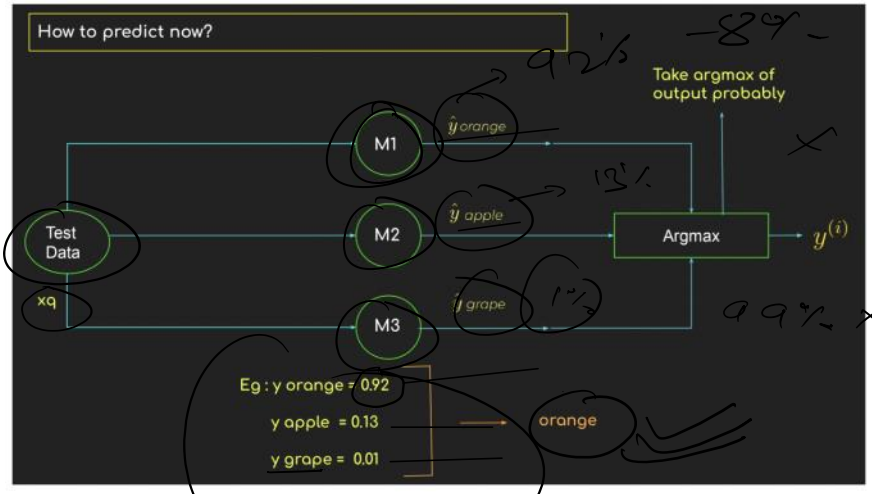
M_2 → apple (1) or not (0)

M_3 → grapes (1) or not (0)



1 → Orange
2 → Apple
3 → grapes

→ Train 3 models



[0.92, 0.13, 0.01]

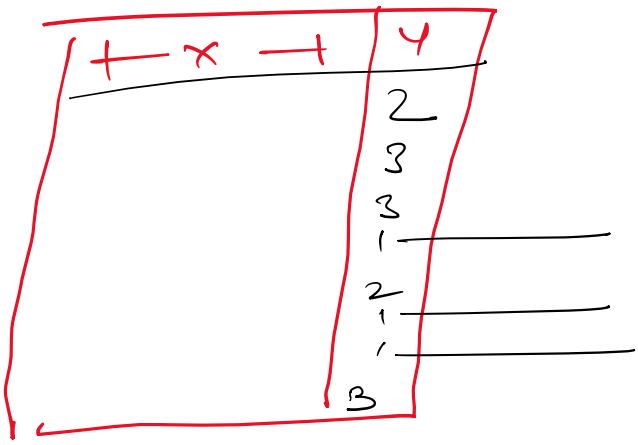
Orang

Orange

max prob

1 → orange
2 → apple

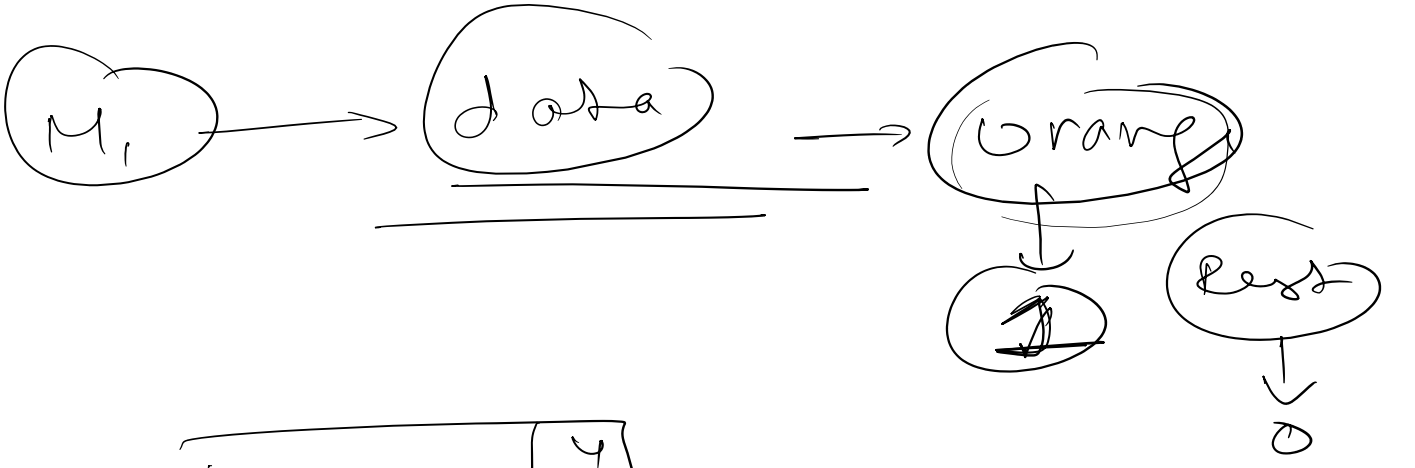


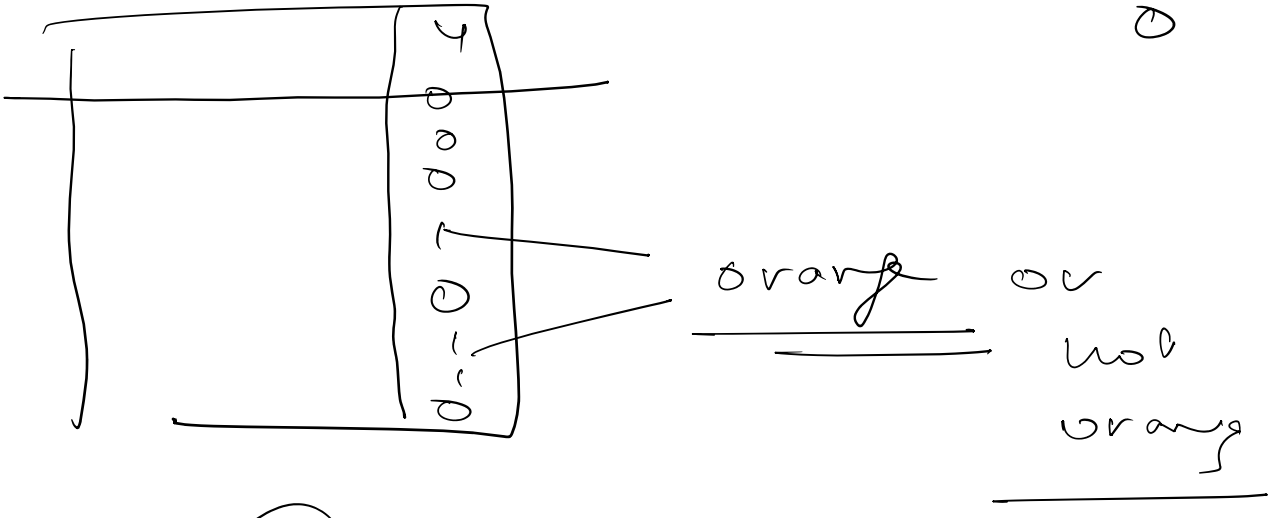


2 → apple

3 → oranges

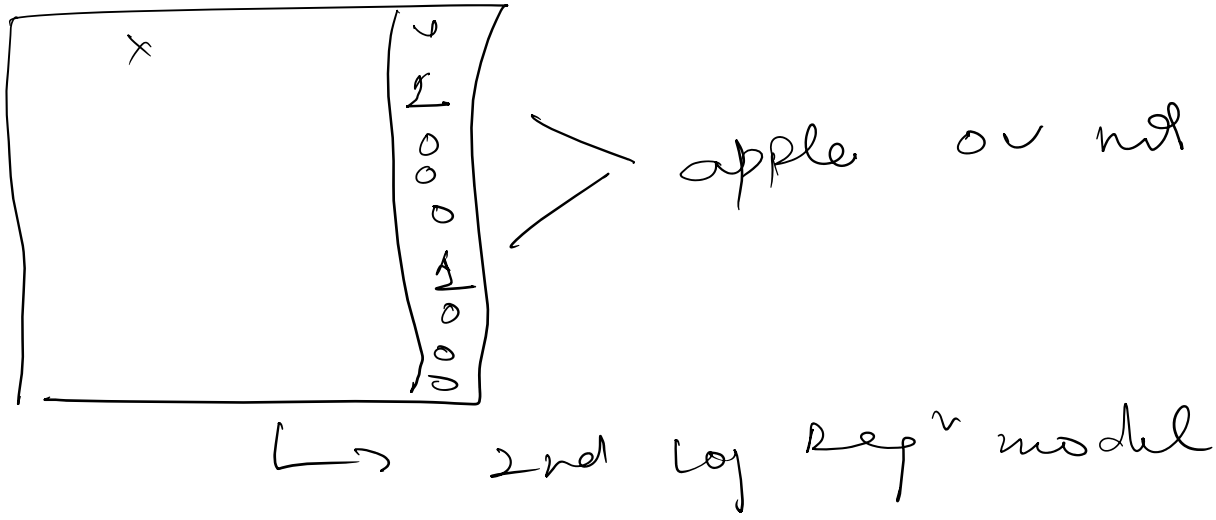
3 models → Binary Class models





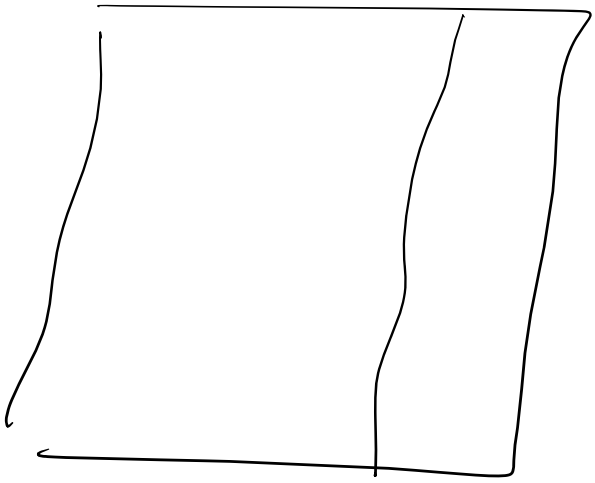
M2 → Log Reg"


M_2
data
apple
2
1
Rest
0


x
1
0
0
0
1
0
0
0
apple or not
2nd log Reg^n model

Logp

4



→ grapes : or not

